

3,500 ft Rocket — Materials Guide

*Recommended material for every physical component, matched to a frugal, reusable G-motor build.
Reference for Step 2 (ordering) and Step 7 (assembly) of the project plan.*

Overview Table

Quick reference for every component. Details and reasoning for each follow below.

Component	Recommended Material	Why	Alternative
Nose cone	Injection-molded plastic (ABS or PVC)	Durable through repeated hard landings; lightweight; clean RF window for GPS	Hand-shaped balsa (cheaper, more fragile, needs sealing)
Body tube	Kraft-phenolic wound paper tube	Stiffer & more heat/scorch resistant than cardstock; far lighter than fiberglass	Plain cardstock (cheaper, weaker); fiberglass (overkill weight for G motors)
Coupler	Same phenolic tube stock (one size down) or dedicated coupler tube	Matches body tube properties; simple friction-fit joint	Plywood ring coupler for a more rigid joint
Motor mount tube	Phenolic or kraft paper motor tube (29 mm standard)	Handles motor casing heat; standard off-the-shelf size	None needed — this is already the standard part
Centering rings	Aircraft-grade birch plywood, 1/8 inch (3.2 mm)	Rigid, light, glues cleanly to both tube and motor mount	Fiberglass rings (more durable, unnecessary at G-motor stress levels)
Fins	Aircraft-grade birch plywood, 1/8 inch (3.2 mm)	Survives hard landings, unlike balsa — matches reusability goal	Basswood (lighter, less durable); balsa (lightest, least durable)
Bulkhead	Birch plywood, 1/8 inch (3.2 mm)	Structural anchor point for shock cord; needs to hold real load	Dense hardwood disk (heavier, unnecessary)
Motor retainer	Aftermarket screw-on plastic or aluminum retainer	Reliable positive retention; friction-fit alone risks losing the casing on ejection	Masking tape retention (not recommended — failure-prone)
Shock cord	Tubular nylon or flat kevlar webbing	Handles ejection-charge shock loads without stretching to failure	Elastic/bungee cord (only for very small, light builds)

Component	Recommended Material	Why	Alternative
Recovery: parachute	Ripstop nylon	Lightweight, packs small, standard hobby recovery material	Plastic sheet streamer (cheaper, less controlled descent)
Recovery: heat protection	Nomex or other fireproof cloth blanket	Protects the parachute from the ejection charge's hot gases	Wadding/dog barf (cellulose wadding) as a consumable alternative
Launch lugs	Plastic or phenolic tube stock	Low friction against the launch rod; negligible weight	Brass or aluminum tube (heavier, unnecessary)
Electronics sled	Basswood, balsa, or 3D-printed plastic rail	Light, easy to shape, and easy to remove/reuse between launches	Plywood sled (more durable, small weight penalty)
External finish	Sanding sealer + primer + enamel or acrylic paint	Seals porous tube/fin surfaces; thin coats keep weight down	Unfinished/unsealed (lighter, but weaker and less weather-resistant)

Detail: Airframe Materials

Body Tube — Kraft-Phenolic Wound Paper

This is the genuine standard at G-motor power, not a compromise pick. Phenolic tubing is stiffer and far more heat- and scorch-resistant than basic cardstock, which matters near the motor mount and ejection charge. At the same time it is much lighter and cheaper than fiberglass, which is unnecessary weight for the impulse levels a G motor produces. Look for 2.6 inch (66 millimeter) phenolic tubing from LOC Precision, Always Ready Rocketry, or Apogee Rockets' house-brand stock. Typical cost: \$8–15 for a few feet.

Nose Cone — Injection-Molded Plastic (ABS or PVC)

Plastic nose cones sized for 2.6 inch (66 millimeter) tubes are stock items at LOC Precision, Public Missiles Ltd, and Apogee. Plastic survives repeated hard landings that would crack or split a balsa nose cone — directly supporting the reusability goal for the airframe, not just the electronics. It is also lightweight and gives the onboard GPS a clean, non-metallic housing near the top of the rocket for signal reception. A hand-shaped balsa nose cone is a cheaper alternative but is more fragile and needs its porous surface sealed before flight.

Fins — Aircraft-Grade Birch Plywood, 1/8 inch (3.2 mm)

Void-free aircraft/craft-grade birch plywood (not construction-grade) is sold specifically as fin stock by Apogee and LOC Precision, and is also available at general hobby or craft stores. Balsa is lighter but dents and snaps on hard landings; since the goal is an airframe (not just electronics) that survives multiple launches, plywood is the more realistic choice. Basswood is a reasonable middle ground — lighter than plywood, tougher than balsa — worth considering if shaving weight matters more than maximum durability. Sand fin edges to a rounded or airfoil profile to cut drag, then seal with 2–3 coats of sanding sealer before paint.

Centering Rings & Bulkhead — Birch Plywood, 1/8 inch (3.2 mm)

The same plywood used for fins works well here. Centering rings need to be rigid enough to hold the motor mount tube perfectly concentric inside the body tube under thrust loads, and the bulkhead needs to hold the recovery harness's shock-cord anchor point under the sharp jolt of ejection — both are structural loads plywood handles comfortably at G-motor scale without adding unnecessary mass.

Detail: Recovery System Materials

Parachute — Ripstop Nylon

Ripstop nylon is the standard hobby-rocketry recovery material: lightweight, packs down small, and resists tearing if a small puncture occurs. Size the canopy for a 15–20 feet per second (4.6–6.1 meters per second) descent rate given the rocket's total loaded weight — slow enough to protect the reusable electronics sled on landing, fast enough to avoid drifting far off the recovery area.

Shock Cord — Tubular Nylon or Flat Kevlar Webbing

The shock cord absorbs the sharp jolt of ejection-charge deployment. Tubular nylon or flat kevlar webbing handles that load without stretching to failure the way a plain elastic or bungee cord can on anything beyond a very small, light rocket. Anchor it to a bulkhead or the motor mount with a U-bolt or epoxied loop, not tape alone.

Heat Protection — Nomex or Fireproof Cloth

A Nomex or other fireproof cloth blanket, layered between the ejection charge and the parachute, prevents the hot gases from the charge from melting or scorching the chute material. Cellulose wadding (sometimes called “dog barf”) is a cheaper, consumable alternative used by many hobbyists, though it needs to be replaced every flight rather than reused.

Detail: Electronics & Hardware

Electronics Sled — Basswood, Balsa, or 3D-Printed Rail

The sled itself can be lighter-duty than the airframe, since it lives inside the protected body tube rather than absorbing landing impacts directly. Basswood or balsa is easy to cut and shape to fit your specific flight computer, GPS, radio, and battery stack. A 3D-printed rail or tray is a durable alternative if a printer is available, and holds up to repeated insertion and removal better than wood — worth it given the sled is meant to be reused across every future build.

Motor Retainer — Aftermarket Screw-On Plastic or Aluminum

A dedicated 29 millimeter motor retainer (widely available from Apogee and similar vendors) threads onto the motor mount tube and holds the spent motor casing in place through ejection. This is worth buying rather than improvising — friction-fit or taped retention risks the spent casing being blown out of the mount by the ejection charge.

Launch Lugs — Plastic or Phenolic Tube Stock

Small-diameter plastic or phenolic tube segments, epoxied to the outside of the body tube, guide the rocket along the launch rod during the first few feet of flight before the fins alone can stabilize it. Two lugs, aligned with each other along the same side of the airframe and clear of all fins, are standard practice.

Finish — Sanding Sealer, Primer, and Thin Paint Coats

Phenolic tube and plywood fins both benefit from a sealed, primed surface before painting — it protects against moisture and minor handling wear between launches. Keep every coat thin: paint adds weight that comes directly out of the altitude budget, so a light, functional finish is preferable to a heavy show finish on a rocket built to reach a specific target apogee.